

(bars: lengths by R , gain by a_{nom} ; hats on estimates; $e_\theta = \theta - \hat{\theta}$; posterior quantities unless marked $-$; $\bar{w}_d[k]$ commanded height, $\Delta\bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$; $b := b_{\text{true}}(\bar{w}[k-1])$ exogenous, slot 5 locked; $\bar{a}' := b(1 - \hat{a}_w[k])^2$, $\bar{a}'' := -2b(1 - \hat{a}_w[k])\bar{a}'$, both read at the estimated gain; $\frac{\partial\bar{a}'}{\partial\hat{a}_w} = -2b(1 - \hat{a}_w[k]) = \bar{a}''/\bar{a}'$; $\Delta\bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$; $\Delta\hat{w}[k] := \Delta\bar{w}_d[k] + (1 - \lambda_c)[\delta\hat{w}_3[k] + y_1[k] - \delta\hat{w}_1[k]]$; $e_{\Delta\bar{w}} := \Delta\bar{w} - \Delta\hat{w}$; $\mathbb{E}[\cdot]$ mean over noise realizations conditional on the data up to k ; $P_{ij} = \mathbb{E}[e_i e_j]$; $d = 2$)

1 True dynamics

$$\begin{aligned}\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\ \delta\bar{w}_3[k+1] &= \lambda_c \delta\bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\ \bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k] + \frac{1}{2} \bar{a}''_{\text{true}}(\bar{w}[k]) \Delta\bar{w}[k]^2 + \mathcal{O}(\Delta\bar{w}^3) \\ \bar{a}'_{\text{true}}(\bar{w}[k]) &= b(1 - \bar{a}_w[k])^2 = b(1 - \hat{a}_w[k] - e_{\bar{a}_w}[k])^2 = \bar{a}' + \frac{\partial\bar{a}'}{\partial\hat{a}_w} e_{\bar{a}_w}[k] + \mathcal{O}(e^2) \\ \bar{a}''_{\text{true}}(\bar{w}[k]) &= -2b(1 - \bar{a}_w[k]) \bar{a}'_{\text{true}}(\bar{w}[k]) = \bar{a}'' + \mathcal{O}(e)\end{aligned}$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \left(\bar{a}' + \frac{\partial\bar{a}'}{\partial\hat{a}_w} e_{\bar{a}_w}[k]\right) \Delta\bar{w}[k] + \frac{1}{2} \bar{a}'' \Delta\bar{w}[k]^2$$

$$\begin{aligned}F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\ \varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] + (1 - \lambda_c) n_w[k] \\ y_1[k] &= \delta\bar{w}_1[k] + n_w[k]\end{aligned}$$

$$\begin{aligned}\mathbb{E}[\varepsilon_w[k] n_w[k]] &= (1 - \lambda_c) R_1 \\ \tilde{\varepsilon}_w[k] &:= \varepsilon_w[k] - (1 - \lambda_c) n_w[k], \quad \mathbb{E}[\tilde{\varepsilon}_w n_w] = 0, \quad \mathbb{E}[\tilde{\varepsilon}_w^2] = Q_{33} - (1 - \lambda_c)^2 R_1 =: \tilde{Q}_{33} \\ \Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) [\delta\bar{w}_3[k] + n_w[k]] + F_{dw}[k] e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]\end{aligned}$$

2 Estimator

$$\begin{aligned}\delta\hat{w}_3[k+1] &= \lambda_c \delta\hat{w}_3[k] - (1 - \lambda_c) [y_1[k] - \delta\hat{w}_1[k]] \\ (\bar{w}_d[k+1] - \delta\hat{w}_3[k+1]) - (\bar{w}_d[k] - \delta\hat{w}_3[k]) &= \Delta\hat{w}[k] \\ \Delta\hat{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c) [\delta\hat{w}_3[k] + y_1[k] - \delta\hat{w}_1[k]] \\ \hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta\hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta\hat{w}[k]^2\end{aligned}$$

3 Error increment

$$y_1[k] - \delta \hat{w}_1[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$\Delta \bar{w}[k] = \Delta \hat{w}[k] + e_{\Delta \bar{w}}[k]$$

$$e_{\Delta \bar{w}}[k] = (1 - \lambda_c) [e_{\delta \bar{w}_3}[k] - e_{\delta \bar{w}_1}[k]] + F_{dw}[k] e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]$$

$$\begin{aligned} e_{\bar{a}_w}[k+1] &= e_{\bar{a}_w}[k] + \left(\bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \right) \Delta \bar{w} + \frac{1}{2} \bar{a}'' \Delta \bar{w}^2 - \bar{a}' \Delta \hat{w} - \frac{1}{2} \bar{a}'' \Delta \hat{w}^2 \\ \left(\bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \right) \Delta \bar{w} &= \left(\bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \right) (\Delta \hat{w} + e_{\Delta \bar{w}}) = \bar{a}' \Delta \hat{w} + \bar{a}' e_{\Delta \bar{w}} + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \Delta \hat{w} + \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} e_{\Delta \bar{w}} \\ \frac{1}{2} \bar{a}'' \Delta \bar{w}^2 &= \frac{1}{2} \bar{a}'' (\Delta \hat{w} + e_{\Delta \bar{w}})^2 = \frac{1}{2} \bar{a}'' \Delta \hat{w}^2 + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 \end{aligned}$$

$$\begin{aligned} e_{\bar{a}_w}[k+1] &= e_{\bar{a}_w}[k] \\ &+ \bar{a}' e_{\Delta \bar{w}} \end{aligned} \tag{1}$$

$$+ \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} \Delta \hat{w} \tag{2}$$

$$+ \frac{\partial \bar{a}'}{\partial \hat{a}_w} e_{\bar{a}_w} e_{\Delta \bar{w}} \tag{3}$$

$$+ \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} \tag{4}$$

$$+ \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 + \mathcal{O}(3) \tag{5}$$

$(\bar{a}' \Delta \hat{w})$ and $\frac{1}{2} \bar{a}'' \Delta \hat{w}^2$ cancel between the true and the estimated step; (1)–(5) name the five terms on those lines; (2) and (3) carry $e_{\bar{a}_w}$ where the $\bar{a}'_{\text{true}}$ copy carries $-\bar{a}'' e_{\delta \bar{w}_3}$

4 Mean

$$\mathbb{E}[e_{\delta \bar{w}_1}] = \mathbb{E}[e_{\delta \bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w] = 0$$

$$\mathbb{E}[e_{\delta \bar{w}_3}^2] = P_{33}, \quad \mathbb{E}[e_{\delta \bar{w}_1}^2] = P_{11}, \quad \mathbb{E}[e_{\delta \bar{w}_1} e_{\delta \bar{w}_3}] = P_{13}$$

$$\mathbb{E}[e_{\delta \bar{w}_3} e_{\bar{a}_w}] = P_{34}, \quad \mathbb{E}[e_{\delta \bar{w}_1} e_{\bar{a}_w}] = P_{14}, \quad \mathbb{E}[e_{\bar{a}_w}^2] = P_{44}$$

$$\mathbb{E}[\tilde{\varepsilon}_w e_{\delta \bar{w}_1}] = \mathbb{E}[\tilde{\varepsilon}_w e_{\delta \bar{w}_3}] = \mathbb{E}[\tilde{\varepsilon}_w e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w n_w] = 0$$

$$\mathbb{E}[e_j \delta \hat{w}_i] = \mathbb{E}[e_j \hat{a}_w] = \mathbb{E}[e_j y_1] = 0 \quad (\text{posterior error} \perp \text{data})$$

$$\mathbb{E}[e_{\bar{a}_w} e_{\Delta \bar{w}}] = (1 - \lambda_c)(P_{34} - P_{14}) + F_{dw} P_{44}$$

$$\mathbb{E}[e_{\Delta \bar{w}}^2] = (1 - \lambda_c)^2(P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c)F_{dw}(P_{34} - P_{14}) + F_{dw}^2 P_{44} + \tilde{Q}_{33}$$

$$\mathbb{E}[(1)] = \bar{a}' \mathbb{E}[e_{\Delta\bar{w}}] = 0$$

$$\mathbb{E}[(2)] = \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w} \mathbb{E}[e_{\bar{a}_w}] = 0$$

$$\mathbb{E}[(3)] = \frac{\partial \bar{a}'}{\partial \hat{a}_w} \mathbb{E}[e_{\bar{a}_w} e_{\Delta\bar{w}}] = \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} - P_{14}) + F_{dw} P_{44}]$$

$$\mathbb{E}[(4)] = \bar{a}'' \Delta \hat{w} \mathbb{E}[e_{\Delta\bar{w}}] = 0$$

$$\begin{aligned} \mathbb{E}[(5)] = \frac{1}{2} \bar{a}'' \mathbb{E}[e_{\Delta\bar{w}}^2] &= \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c) F_{dw} (P_{34} - P_{14}) \\ &\quad + F_{dw}^2 P_{44} + \tilde{Q}_{33}] \end{aligned}$$

$$\begin{aligned} \mathbb{E}[e_{\bar{a}_w}[k+1]] &= \mathbb{E}[e_{\bar{a}_w}[k]] + \mathbb{E}[(3)] + \mathbb{E}[(5)] \\ &= \mathbb{E}[e_{\bar{a}_w}[k]] + \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} - P_{14}) + F_{dw} P_{44}] \\ &\quad + \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c) F_{dw} (P_{34} - P_{14}) + F_{dw}^2 P_{44} + \tilde{Q}_{33}] \end{aligned}$$

$$\begin{aligned} \mathbb{E}[(3)] - \mathbb{E}[(3)]_{a'_{\text{true}}} &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} - P_{14}) + F_{dw} P_{44}] + \bar{a}'' [(1 - \lambda_c)(P_{33} - P_{13}) + F_{dw} P_{34}] \\ &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34} + \bar{a}' P_{33} - P_{14} - \bar{a}' P_{13}) + F_{dw} (P_{44} + \bar{a}' P_{34})] \\ &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} \mathbb{E}[(e_{\bar{a}_w} + \bar{a}' e_{\delta\bar{w}_3}) e_{\Delta\bar{w}}] \end{aligned}$$

((1), (2), (4): $\bar{a}'$ and $\Delta \hat{w}$ are functions of the data, $e_{\Delta\bar{w}}$ and $e_{\bar{a}_w}$ are posterior errors, $\mathbb{E}[e \mid \text{data}] = 0$; without the $y_1 - \delta \hat{w}_1$ input $e_{\Delta\bar{w}}$ contains $(1 - \lambda_c)n_w$, correlated with the data, and these three means do not vanish; last block: the difference to the a'_{true} copy's start-point line, using $\bar{a}'' = \frac{\partial \bar{a}'}{\partial \hat{a}_w} \bar{a}'$; it is zero when $e_{\bar{a}_w} = -\bar{a}' e_{\delta\bar{w}_3}$, i.e. $P_{34} = -\bar{a}' P_{33}$, $P_{14} = -\bar{a}' P_{13}$, $P_{44} = \bar{a}'^2 P_{33}$)

5 Estimator with the mean term

$$\begin{aligned} \hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta \hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta \hat{w}[k]^2 \\ &\quad + \frac{\partial \bar{a}'}{\partial \hat{a}_w} [(1 - \lambda_c)(P_{34}[k] - P_{14}[k]) + F_{dw}[k] P_{44}[k]] \\ &\quad + \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2 (P_{33}[k] + P_{11}[k] - 2P_{13}[k]) + 2(1 - \lambda_c) F_{dw}[k] (P_{34}[k] - P_{14}[k]) \\ &\quad + F_{dw}[k]^2 P_{44}[k] + \tilde{Q}_{33}[k]] \end{aligned}$$

$$e_{\bar{a}_w}[k+1] = e_{\bar{a}_w}[k] + (1) + (2) + (3) + (4) + (5) - \mathbb{E}[(3)] - \mathbb{E}[(5)]$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] = \mathbb{E}[e_{\bar{a}_w}[k]]$$

(the two added lines are deterministic inputs to the predict; they are not differentiated and do not enter F_e , H , Q)

6 Jacobians

$$\begin{aligned}
\frac{\partial \delta \hat{w}_3[k+1]}{\partial \delta \hat{w}_3[k]} &= \lambda_c, & \frac{\partial \delta \hat{w}_3[k+1]}{\partial \delta \hat{w}_1[k]} &= (1 - \lambda_c) \\
\frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_3[k]} &= (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow (1 - \lambda_c) \bar{a}' \\
\frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_1[k]} &= -(1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow -(1 - \lambda_c) \bar{a}' \\
\frac{\partial \hat{a}_w[k+1]}{\partial \hat{a}_w[k]} &= 1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w}[k] + \frac{1}{2} \frac{\partial \bar{a}''}{\partial \hat{a}_w} \Delta \hat{w}[k]^2 \rightarrow 1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w}[k] \\
\frac{\partial \bar{a}_w[k+1]}{\partial \bar{a}_w[k]} &= 1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \bar{w}[k] + F_{dw}[k](\bar{a}' + \bar{a}'' \Delta \bar{w}[k]) \rightarrow 1 + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w}[k] + F_{dw}[k] \bar{a}'
\end{aligned}$$

(limits at $\Delta \hat{w} \rightarrow 0$ except the $\frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w}$ entry, which the code keeps: the law's own amplification of a gain error along the step; no $-\bar{a}'' \Delta \hat{w}$ in the $\delta \hat{w}_3$ entry, the slope reads a gain, not a height; the last line and $\partial \delta \hat{w}_3[k+1]/\partial \bar{a}_w[k] = -F_{dw}$ are entries of the TRUE transition, not of the estimator's lines: the truth's $F_{dw} e_{\bar{a}_w}$ inside $\Delta \bar{w}$, written as $F_{dw}(\bar{a}_w - \hat{a}_w)$)

7 Error dynamics

$$\begin{aligned}
e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\
e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\
e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] + (1 - \lambda_c) e_{\delta \bar{w}_1}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \tilde{\varepsilon}_w[k] \\
(1) = \bar{a}' e_{\Delta \bar{w}}[k] &= -(1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_1}[k] + (1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_3}[k] + F_{dw}[k] \bar{a}' e_{\bar{a}_w}[k] + \bar{a}' \tilde{\varepsilon}_w[k] \\
(2) &= \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w}[k] e_{\bar{a}_w}[k] \\
e_{\bar{a}_w}[k+1] &= -(1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_1}[k] + (1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_3}[k] + \left(1 + F_{dw} \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w}\right) e_{\bar{a}_w}[k] + \bar{a}' \tilde{\varepsilon}_w[k] \\
&\quad + (3) + (4) + (5) - \mathbb{E}[(3)] - \mathbb{E}[(5)]
\end{aligned}$$

(third line: $\delta \bar{w}_3[k+1] - \delta \hat{w}_3[k+1]$ with $y_1 - \delta \hat{w}_1 = e_{\delta \bar{w}_1} + n_w$, the n_w of ε_w cancels; sixth line: $e_{\bar{a}_w}[k] + (1) + (2)$ gives the first-order coefficients of the matrix below, (2) being first order in $e_{\bar{a}_w}$ with a data coefficient; (3)–(5) minus their mean is the zero-mean remainder)

$$\mathbf{e}[k+1] = \tilde{F}_e[k] \mathbf{e}[k] + \tilde{\mathbf{q}}[k], \quad \tilde{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (1 - \lambda_c) & 0 & \lambda_c & -F_{dw} \\ -(1 - \lambda_c) \bar{a}' & 0 & (1 - \lambda_c) \bar{a}' & 1 + F_{dw} \bar{a}' + \frac{\partial \bar{a}'}{\partial \hat{a}_w} \Delta \hat{w} \end{bmatrix}$$

$$\tilde{F}_e[k] = F_e[k] - g_n[k] \mathbf{e}_1^\top$$

$$g_n[k] = (1 - \lambda_c) \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \quad \tilde{\mathbf{q}}[k] = \tilde{\varepsilon}_w[k] \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \quad \mathbb{E}[\tilde{\mathbf{q}}[k] n_w[k]] = 0$$

$$P^f[k+1] = \tilde{F}_e[k] P[k] \tilde{F}_e[k]^\top + \tilde{Q}[k]$$

8 Measurement and update

$$\begin{aligned} \nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k], \quad \bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k] \\ H_{24}[k] &= \frac{\partial (\hat{a}_w - \bar{a}' \nabla_d \bar{w}_d)}{\partial \hat{a}_w} = 1 - \frac{\partial \bar{a}'}{\partial \hat{a}_w} \nabla_d \bar{w}_d[k] \\ H &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{24} \end{bmatrix} \\ e_{y_1}[k] &= e_{\delta \bar{w}_1}^-[k] + n_w[k], \quad e_{y_2}[k] = H_{24}[k] e_{\bar{a}_w}^-[k] + n_a[k] \end{aligned}$$

$$\begin{aligned} L[k] &= P^f H^\top (H P^f H^\top + R)^{-1} \\ \hat{\mathbf{x}}[k] &= \hat{\mathbf{x}}^-[k] + L[k] \mathbf{e}_y[k] \\ P[k] &= (I - L[k] H) P^f[k] \\ y_1[k] - \delta \hat{w}_1[k] &= \frac{R_1}{P_{11}^f[k] + R_1} e_{y_1}[k] - \ell_{12}[k] (e_{y_2}[k] - \ell_{41}[k] e_{y_1}[k]) = \mathbb{E}[n_w[k] \mid y[1..k]] \end{aligned}$$

($H_{24} \neq 1$ because $\bar{a}'$ reads $\hat{a}_w$; last line: the posterior residual the Estimator section feeds into the next step, written for the sequential update y_1 then y_2 as in the code)

9 Noise

$$\begin{aligned} \kappa_T &= \frac{4k_B T a_o}{R}, \quad \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2} \\ \tilde{Q}_{33} &= \kappa_T (\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2])) \quad (= Q_{33} - (1 - \lambda_c)^2 \sigma_{n_w}^2) \\ \tilde{Q}_{34} &= \tilde{Q}_{43} = -\bar{a}' \tilde{Q}_{33} \\ \tilde{Q}_{44} &= \bar{a}'^2 \tilde{Q}_{33} \\ R &= \text{diag}(R_1, R_2), \quad R_1 = \sigma_{n_w}^2, \quad R_2 = 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w)(\hat{a}_w + \xi)^2 \end{aligned}$$

(IF_{var} , ξ as in 0902_formC_aptrue_4state.tex; the n_w share $(1 - \lambda_c)^2 \sigma_{n_w}^2$ has left Q and entered the Estimator section as the input $g_n(y_1 - \delta \hat{w}_1)$)



